Quantum Mechanics II — Exercise Sheet 11

Scattering amplitudes and flux

Variant: Questions only

Exercise 1 (A downward step). A particle of energy $E > 0$ meets $V(x) = 0$ for $x < 0$ and $V(x) = -V_0$ for $x > 0$, with $V_0 > 0$. For unit incident amplitude from the left, match the wave and derivative at zero. (i) Find r, t, R and T. (ii) Explain why the transmitted amplitude need not be the square root of the transmission probability.

Exercise 2 (Two delta barriers). Consider $V(x) = \alpha\delta(x) + \alpha\delta(x - d)$, with $\alpha > 0$ and $d > 0$. Take as given the composition formula

$$t_2(k) = \frac{t(k)^2 e^{\mathrm{i}kd}}{1 - r(k)^2 e^{2\mathrm{i}kd}}, \qquad t(k) = \frac{k}{k + \mathrm{i}\beta}, \quad r(k) = \frac{-\mathrm{i}\beta}{k + \mathrm{i}\beta}, \quad \beta = \frac{m\alpha}{\hbar^2}, \quad k > 0.$$

Derive $T_2 = |t_2|^2$ and the condition for perfect transmission. Show that no positive-energy pole occurs in the denominator.

Exercise 3 (Resonances above a rectangular barrier). A barrier of height $V_0 > 0$ occupies $(0, \ell)$, and $E > V_0$. Let $k = \sqrt{2mE}/\hbar$ and $q = \sqrt{2m(E - V_0)}/\hbar$. Match $e^{\mathrm{i}kx} + re^{-\mathrm{i}kx}$ on the left, $Ae^{\mathrm{i}qx} + Be^{-\mathrm{i}qx}$ inside, and $te^{\mathrm{i}kx}$ on the right. Derive $T = |t|^2$ and find every energy of perfect transmission. Compute the limit as $E \downarrow V_0$ and decide whether the zero interior wave number is a perfect-transmission resonance.

Exercise 4 (A packet averages probabilities). A normalized incoming momentum law is uniform on $[k_0, 2k_0]$ in wave number, where $k_0 > 0$. It scatters from a single delta barrier with $\beta = k_0$ and $T(k) = k^2/(k^2 + \beta^2)$. Compute the asymptotic transmission probability and compare it with $T(3k_0/2)$. For this delta model take as an additional given the packet identity

$$P_{\mathrm{trans}} = \int_0^\infty T(k)w(k)\,\mathrm{d}k,$$

where w is the normalized incoming wave-number density. The bounded-potential theorem in Lecture 11 is not being applied outside its hypotheses.